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Conclusion

Possible solutions for the issues identified in previous chapters will be grouped loosely into technology and market solutions, followed by legal and policy solutions, reflecting Lessig's taxonomy of the four regulatory modalities of cyberspace: 'code' (i.e. technology and architecture), market, law and norms.¹

The main idea advanced in this book is that the interests of individual users to decide what happens to their data on death should be recognised and protected. In other words, the book promotes user autonomy, privacy, identity and technological options that correspond with the offline, traditional ones (i.e. freedom of testation). The solutions also recognise novel phenomena of immortality, considered under the concept of postmortal privacy. This is not to say that some other legitimate interests should be disregarded, such as public interest, the need to maintain archives and historical records, freedom of expression or the freedom to conduct business. However, due to the scope of this book, an in-depth examination and balancing of these interests is not viable.

All these solutions are tentative. They originate from the analysis in this book, and I will aim to develop further the principles established here in future research projects and publications. The ambition of this book was to set the scene, introduce and explore critical theoretical and legal issues, and offer a framework for future thinking, rather than to offer precise solutions for each issue.

7.1 'Code' and Market Solutions

'Code' or technology solutions, using Lessig's taxonomy mentioned above, are online services aiming to assist in the disposition of digital assets on death, digital estate planning, succession and remembrance. A significant category of stakeholders is online services, which aim to help with the disposition of digital assets on death. They aim to shift the control of digital assets to users

¹ L Lessig, *Code: Version 2.0* (Basic Books 2006).

by enabling the designation of beneficiaries who will receive the passwords and content of digital asset accounts. This way, the services impliedly recognise user property in digital assets and attempt to bypass restrictions imposed by the terms of service discussed in this book. Understandably, they also see a growing market and an opportunity for investment and profit.

These services, still primarily US-based but increasingly hosted in the UK, are categorised in this book as follows:

- *Digital wills*, for example Legacy Locker, now part of the True Key service,² Cirrus Legacy,³ Beyond Life⁴ or SecureSafe⁵. This kind of service works on the principle that a user stores their account information and passwords, and nominates a 'digital heir' or a beneficiary who will get access to this information. The beneficiary or someone else (e.g. Account Activator in SecureSafe) reports their death and provides a death certificate, and after a validation process the digital heir gets access to the account information. The SecureSafe service has now changed its business model entirely, focusing on password management for businesses.
- *Messaging services*, for example ifidie, a Facebook application that allows users to leave messages to Facebook friends on death.⁶
- *Memorial and legacy websites*, for example BCelebrated, an autobiographical legacy website that offers a personalised memorial website, protected private messages and emails, activated by chosen activators.⁷
- *Combined solutions*, combining offline and online services. For example, My Digital Executor, which involves two solicitors who keep email accounts, passwords and a list of digital assets. Assets are then transferred in the form of codicil;⁸ MyWishes, which offers a comprehensive list of services, including digital estate planning, will writing, bucket list and advance care planning;⁹ and Final Fling, which provides advice on living

² Password Box, 'Legacy Locker' (now True Key) <<https://www.passwordbox.com/legacy-locker>> last accessed 1 August 2021.

³ Cirrus Legacy, 'Making it easy for your guardians' <<http://www.cirruslegacy.com/139-guardians.html>> last accessed 15 June 2019 (no longer available).

⁴ Beyond Life <<https://beyond.life/make-will-online>> last accessed 1 August 2021.

⁵ SecureSafe, 'Questions about data inheritance' <<http://www.securesafe.com/en/faq/inheritance/>> last accessed 1 August 2021 (now changed its business model).

⁶ ifidie, 'What happens to your Facebook profile if you die?' <<http://ifidie.net>> last accessed 1 August 2021.

⁷ BCelebrated <<http://www.bcelebrated.com/>> last accessed 1 August 2021.

⁸ My Digital Executor <<http://www.mydigitalexecutor.co.uk/the-solution-2/>> last accessed 1 August 2021.

⁹ MyWishes <<https://www.mywishes.co.uk/how-it-works>> last accessed 1 August 2021.

and dying well (funerals, celebrations, bereavement, treasure trove, wills, advance decision, safe deposit box and essential documents; wills printed out, signed and witnessed).¹⁰

Lamm et al. categorise these solutions differently, focusing on the character of actions they promise to undertake on death. Accordingly, they find four categories: services offering to store passwords; services facilitating the administration of digital assets; services performing specific actions (e.g. removing all the data on behalf of a deceased person); and services that currently do not exist, but hypothetically provide their services through partnerships with the service providers of the deceased's accounts.¹¹ This categorisation is very similar to the one used in this book, with the slight difference that it focuses on actions rather than on business models.

In our earlier work, Edwards and I evaluated some of the 'code' solutions and concluded that 'these are not themselves a foolproof solution'¹² for five main reasons:

- They could cause a breach of terms of service (due to the non-transferable nature of most assets, as suggested in the previous chapters).
- There is a danger of committing a criminal offence (according to the provisions of the anti-interception and privacy laws; see previous chapters).
- The services are inconsistent with the law of succession and executry (they do not fulfil the requirements of will formalities; conflicts may arise with the interests of heirs under wills or laws of intestacy; there are jurisdiction issues, etc.).
- There are concerns over the business viability and longevity of the market (this has proven correct for many services, as indicated above).
- The issues of security and identity theft (the services store passwords and keys to valuable assets and personal data).¹³

Similarly, Beyer and Cahn, and Lamm et al. identify most of these problems.¹⁴

¹⁰ Final Fling <<http://blog.finalfling.com/>> last accessed 1 August 2021.

¹¹ J Lamm et al., 'The digital death conundrum: how federal and state laws prevent fiduciaries from managing digital property' (2014) 68 *University of Miami Law Review* 385, 408.

¹² L Edwards and E Harbinja, "What happens to my Facebook profile when I die?": Legal issues around transmission of digital assets on death' in C Maciel and V Pereira (eds), *Digital Legacy and Interaction: Post-Mortem Issues* (Springer 2013) 144.

¹³ Ibid.

¹⁴ N Cahn, 'Probate law meets the digital age' (2014) 67 *Vanderbilt Law Review* 1697, 1706; Lamm et al. (n 11) 400–1.

The issues are significant, and it is not recommended that the services be used in their current form and with the law as it stands. However, with improvements and provided that the law recognises them, they may be suitable for broader use in the future. In principle, the services are more suitable for the online environment; they recognise the technological features of digital assets and enable an automated transmission on death. Conversely, the issues surrounding them are numerous and complex, so I do not envisage their legitimate reception soon, at least not outside the US, where the RUFADAA might encourage their use.

In addition to these more established code solutions, a recent phenomenon of the deceased's deepfakes and chatbots or ghostbots deserves some consideration. For instance, a MyHeritage service called DeepNostalgia allows users to animate photographs of late relatives and share them across various platforms.¹⁵ The service uses technology licensed from a deep-learning start-up company called D-ID,¹⁶ in which video re-enactment, fixed sequences of movements and gestures are applied to uploaded photos, allowing individuals in the photo to smile and blink.¹⁷ The technology is still in quite a rudimentary form, but it nevertheless raises questions of consent, reputation, personality and identity post-mortem, the issues discussed in this book.

A more advanced version is a chatbot created using AI and the deceased's data. Recently, Jason Rohrer, an indie games developer, created Samantha, a chatbot built using OpenAI's GPT-3 application programming interface (API). The software was then used by thousands of people, including a man who used the program to simulate his late fiancée. Due to a public outcry and the alleged violations of its rules, OpenAI discontinued Rohrer's access to its GPT-3 API.¹⁸ These technologies add to the number of problems identified for code solutions. Currently, these problems are predominantly ethical and revolve around big questions of immortality, recreation and existence, but will soon become more relevant for the law. For example, the EU's recent Proposal for AI Regulations in Article 52 requires users of AI systems that generate or manipulate 'image, audio or video content that appreciably resembles existing persons, objects, places or other entities or events and would falsely appear to a person to be authentic or truthful ("deep fake")' to 'disclose that

¹⁵ MyHeritage <<https://www.myheritage.com/deep-nostalgia>> last accessed 1 August 2021.

¹⁶ D-ID <<https://www.d-id.com/>> last accessed 1 August 2021.

¹⁷ MyHeritage, 'FAQ' in MyHeritage (n 15).

¹⁸ K Quach, 'A developer built an AI chatbot using GPT-3 that helped a man speak again to his late fiancée. OpenAI shut it down' *The Register* (8 September 2021) <https://www.theregister.com/2021/09/08/project_december_openai_gpt_3/> last accessed 8 September 2021.

the content has been artificially generated or manipulated'.¹⁹ This provision merely requires transparency and does not answer questions specific to the creation of the deceased's deepfakes and chatbots, such as whether this 'content' infringes the deceased's rights of privacy, autonomy and dignity; who has the right to use such a system and create deepfakes; what dataset is being used; and whether consent is necessary. Understandably, the proposal does not aim to address all these issues either, so a more holistic law reform is necessary. This is a developing area that goes beyond the questions examined in this book, and I will continue exploring it in my future research.²⁰ I have mentioned it here for completeness.

(c) In-Service Solutions

For the purpose of this book, in-service solutions include technology and service features that enable the transmission of a digital asset or part of the deceased's content to beneficiaries in a service internally. I have analysed Google Inactive Account Manager and Facebook Legacy Contact extensively in Chapters 4 and 6. These solutions have been implemented mainly in response to pressure placed on service providers by the media and families of deceased users, rather than as a result of pressure from users themselves or being initiated by providers. The likely reason is the general reluctance of individuals to contemplate their death, and providers' fear that reminders of it would damage their reputations. However, the market is moving towards providing answers, and these efforts should be recognised and supported by policy and the law.

Notably, the existing solutions for the transmission of digital assets within service providers should evolve. Service providers need to make them more visible because they are barely used and not advertised. Service providers should amend their terms of service and make them more coherent with their post-mortem solutions. For instance, they should recognise the transferability of content on death explicitly in their terms of service.

In virtual worlds, these terms will allow personal representatives of a deceased user to access and recoup any benefits arising from virtual world usufruct. Thus, a personal representative could sell valuable items a user left behind, for example through online auctions, and pass the monetary value to the user's heirs or beneficiaries.

¹⁹ Proposal for a Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence act) and amending certain union legislative acts, Brussels, 21.4.2021, COM(2021) 206 final, 2021/0106(COD).

²⁰ E Harbinja, L Edwards and M McVey, 'Governing ghostbots' (forthcoming).

Email providers should aim to make the provisions of their contracts with users more coherent (those relating to ownership and the transfer of a user's content). In addition, given the prevalence of post-mortem privacy, they should recognise this phenomenon more clearly in their terms of service. The user should decide on the deletion of the content or leaving some of it to a beneficiary. The visibility of these options is vital, and the user should be adequately informed through terms of service and a 'wizard' or a pop-up window that would facilitate a user's choice.

Facebook's terms of service should explicitly state that a user's choice of Legacy Contact trumps the option of memorialisation and requests submitted by the next of kin or friends. Other social networks, including Facebook's own Instagram, should follow Facebook's lead, in principle, enabling their users to decide whether they wish their account to be deleted on death or if they would like to leave some of their content to a beneficiary. This option should be visible, and users could, for instance, be required to go through a 'wizard' to check their post-mortem choice. At the moment, users are advised to use available post-mortem options, but they should be aware of conflicts that these may cause between the in-service beneficiaries and heirs. For instance, if they leave a piece of content protected by copyright to someone other than their heir, the heir would have a claim against that person. I have explored this in more detail in Chapter 4.

Concerning the suggested deletion of a user's content, it is worth noting that it does not mean deletion of all personal data and content associated with the deceased person, as this is, arguably, impossible. Instead, the deletion would be limited to a particular service and the user's account. Thus, in order to respect the rights of other users in their content, the deletion would not include, for instance, emails in another user's inbox or private messages sent on Facebook. That would be impractical and unfair to the recipient of the communication or content.

In the future, it would be helpful to consider the implementation of Schafer's idea of ZombAI, whereby an intelligent agent (AI) would be used to interpret the deceased's wishes and continue their existence in some form.²¹ For example, the agent could learn from an individual and provide a contextual interpretation of their wishes after death. It could include a tool for the technologically appropriate disposition of digital assets and the post-mortem protection of the deceased's personality and privacy interests. The agent could serve as a 'defender' of the deceased's personality and identity-related interests

²¹ B Schafer, 'On living and undead wills: ZombAIs, technology and the future of inheritance law' in L Edwards, B Schafer and E Harbinja (eds), *Future Law: Emerging Technology, Regulation and Ethics* (Edinburgh University Press 2020).

for a limited time (thirty or fifty years), and it could even adapt to changed circumstances after the individual's death.

The AI agent, developed with ethical, social and legal concerns in mind, would have the capacity to protect postmortem privacy, a novel concept developed in this book. Ideally, this protection would include the totality of the deceased's post-mortem persona, their informational body and digital remains. It would have the potential to address novel concerns around ghostbots and digital resurrection mentioned above. This solution would challenge some core paradigms of succession law, legal personhood, and other traditional areas of law and ethics, so it requires in-depth research and law reform that may not happen very soon.

7.2 Law and Policy Solutions

I reject the proposition that we should leave the regulation of the transmission of digital assets on death to the market. The analysis in this book demonstrates problems for users and society created by new technologies and service providers' contractual conditions. It is argued that legal intervention is required.

We have analysed most of the existing laws that regulate digital assets and post-mortem privacy in Chapter 3. We have identified their deficiencies and argued for a law that would recognise the technological, social and economic reality. No existing law is sufficient *per se*. To be successful, a legislator needs to consider succession law, probate, personality, privacy, data protection, IP, contract, criminal law and other areas considered in this book. Holistic law reform is vital for a phenomenon that goes beyond just assets and wealth and includes digital remains that are highly personal and sensitive. I also agree with Morse and Birnhack, who maintain that users need to be empowered to make decisions about their digital remains, and as such 'a top-down, one-size-fits-all law will frustrate the wishes of large segments of the user population'.²²

An acceptable legal solution should aim to recognise technology as a way of disposing of digital assets, as a more efficient and immediate solution online, one that recognises technological limitations, autonomy and the changing landscape of relationships there (co-constructed profiles and sharing). Succession laws applicable to a traditional estate should not apply to digital assets by default. As concluded in the previous chapters, the content in digital assets is not always property, it is potentially protected by copyright as unpublished works, and it consists mainly of personal data and information.

²² T Morse and M Birnhack, 'The posthumous privacy paradox: Privacy preferences and behavior regarding digital remains' (2020) *New Media and Society* 1, 19.

Further, it includes digital remains that go beyond personal data and privacy and result in immortality, which the law should consider in a comprehensive approach. Therefore, the law should recognise these novel features and enable user choice and transmission of copyrighted material and other content, where applicable. An example of this is the US RUFADAA, analysed in Chapter 3.

In a written submission to the NSW Law Reform Commission, Edwards and I made specific proposals of what the law should ideally include.²³ We argued that access to digital assets should always be exercised after death only by personal representatives (PRs) of the deceased, and there should be no default access for families or friends of the deceased to digital assets. We also maintained that there is a need to balance the heirs' interests in remembering the deceased and receiving economic benefits from them with the privacy interests of the deceased, as well as a need to consider the practical issues for platforms that host these assets. We concluded that the PR is best equipped legally and practically to strike this balance and be a single point of contact for the platform, so it has certainty when giving access to the deceased's assets.

We have identified an issue in jurisdictions that allow automatic access by a PR on death, namely transferring highly private digital content (e.g. intimate blog posts) to family members after death, which may not be what the deceased would have preferred. Accordingly, we suggested that the PR's access rights be amended slightly in relation to certain digital assets to provide for the most nuanced legislative solution to date. Those assets that are intrinsically likely to be highly personal (e.g. friends-locked social media accounts) should be treated differently from those that are likely to be non-private or principally of monetary value (e.g. virtual worlds and games accounts, financial domain names, Amazon or eBay accounts).

For personal assets, we suggest a hierarchy of instructions to be followed by PRs: (a) in-service tools, (b) will, (c) failing this, distribution of assets falls to be governed by any terms of service relevant and only the deceased's catalogue of communications is accessible by default. For monetary assets, unfettered access should be allowed to the PR, and there should be an explicit provision for clarity that this access right trumps platforms' terms of service even if they provide to the contrary. The document required by the PR to secure access should be the same as that required to access typical non-digital assets of the estate, for example a grant of probate of the will or letters of administration.

²³ L Edwards and E Harbinja, 'Written submission', NSW Law Reform Commission: Access to digital assets upon death or incapacity (2019).

These principles should be implemented through amendments to the probate and succession laws in England. This could be effected through a regulation, which would primarily amend the Wills Act 1837 and the Administrations of Estates Act 1925, recognising digital assets as a specific part of the deceased's estate. The Law Commission commenced revising the law of wills in 2017 and this process is expected to last for another few years. Unfortunately, the reform does not include digital assets as the Commission considers these to fall outside its scope.²⁴ Another reform is looking at digital assets, but only considering their property and contract aspects and in so far as they include blockchain and similar technologies.²⁵

Recently, Ian Paisley, a member of the UK Parliament, introduced the Digital Devices (Access for Next of Kin) Bill, proposing to give the next of kin of a dead or incapacitated person default access to devices. It includes access to content a person has bought online and personal data like photographs, emails and messages.²⁶ This *ad hoc*, ill-conceived bill fails to acknowledge all the nuances and complexity discussed in this book. The proposal completely disregards the need to consider various relevant areas of law and construct a holistic law reform, as discussed here. It is to be hoped that Parliament will recognise this and reject the proposal.

A reasonable idea would be to negotiate and recognise these principles at the EU level to harmonise the practice and the law. Recent events suggest some development in this area. In January 2022, the European Commission introduced the European Declaration on Digital Rights and Principles, which reads, 'Everyone should be able to determine their digital legacy, and decide what happens with the publicly available information that concerns them, after their death.'²⁷ The proposal introduces digital legacy more formally in the EU for the first time. However, there are difficulties, as the EU cannot impose solutions on matters of property and substantive succession law, so

²⁴ Law Commission, 'Making a will' (Consultation paper 231, 2017) <<https://www.s3-eu-west-2.amazonaws.com/lawcom-prod-storage-11jsxou24uy7q/uploads/2017/07/Making-a-will-consultation.pdf>> last accessed 1 August 2021.

²⁵ Law Commission, 'Digital assets: Call for evidence' (April 2021) <<https://www.lawcom.gov.uk/project/digital-assets/>> last accessed 1 August 2021.

²⁶ R Bradbury, 'A proposed UK law would automatically hand loved ones access to your messages, photos, and emails after you die' *Business Insider* (31 January 2022) <https://www.businessinsider.com/uk-law-proposed-for-family-get-devices-deceased-2022-1?utm_source=twitter&utm_content=referral&utm_term=topbar&referrer=twitter> last accessed 31 January 2022.

²⁷ European Commission, 'European Declaration on Digital Rights and Principles for the Digital Decade' Brussels, 26.1.2022 COM(2022) 28 final, 5.

the Declaration is likely to remain just a good idea.²⁸ The EU could decide to extend data protection or privacy protections post-mortem, but even that would be a partial solution that disregards a holistic law reform. Even if we ignore this major obstacle, the document includes imprecise terms in its current form, such as ‘determine’ and ‘publicly available information’. The word ‘determine’ should be replaced with ‘control’, and the decision should extend to digital assets and remains, as suggested in this book. Otherwise, the Declaration risks being reductionist and imprecise.

In addition to these general principles for law reform, Chapter 5 introduced a solution distinct from the ones in the other case study chapters. Recognising the conflicting interests of developers and players, in line with the doctrinal and normative analyses of virtual property and the phenomenon of *constitutionalisation* of VWs, the chapter proposes a novel compromise solution in the form of virtual usufruct. This concept pertains to the second level virtual assets and it transmits on death. It is a law reform proposal and could be introduced by terms of service, that is, internally to a VW, or could generally be mandated by the law.

Chapter 6 identified difficulties around the transmission of copyright in unpublished works stored in digital assets. These difficulties can be resolved by explicitly allowing access to a user’s account to a personal representative through the provider’s terms of service, as long as the user has not requested deletion of the account. In addition, copyright law that requires unpublished works to be stored on a medium that would form a part of an estate should be amended and clarified. One of the ways to clarify this is by inserting a provision into the UK Copyright, Designs and Patents Act 1988, which mandates that this provision applies to digital assets and copyright in unpublished works is transmissible (as these are not tangible media or part of an offline estate), provided that the user has not expressed a contrary wish pre-mortem.

It is also suggested that the data protection regime in the UK and the EU should recognise PMP. This can be done, for instance, by envisaging protection of the deceased’s personal data in the member states’ legislation across the EU (akin to what many member states have done; see Chapter 3 for

²⁸ The EU does not interfere with property rights of member states. See Article 345 of the Treaty on the Functioning of the European Union (Consolidated version 2012) OJ C 326, 26.10.2012; Regulation (EU) No 650/2012 of the European Parliament and of the Council of 4 July 2012 on jurisdiction, applicable law, recognition and enforcement of decisions and acceptance and enforcement of authentic instruments in matters of succession and on the creation of a European Certificate of Succession L 201/107. The Regulation deals with cross-border issues in succession and does not interfere with the substantive national succession laws. See for example Recital 15.

more detail). The GDPR permits member states to provide for such protection, but does not protect the deceased's data itself.²⁹ This protection could be time-limited (e.g. thirty or fifty years post-mortem), and it could recognise a user's autonomy and pre-mortem choice (e.g. mandating service providers to require the choice to be expressed during the registration process or on some other appropriate occasion). This suggestion needs to be developed in more detail to address some prominent issues, such as who would consent on behalf of the deceased to process their personal data. Ideally, the EU would pass amendments to the GDPR to this effect, but this is unrealistic given legislative processes and the EU's priorities at the moment. Harmonising this aspect of data protection would contribute to the legislative aim of maximum harmonisation, which has not been achieved concerning the deceased's data. It is also unlikely that the UK Data Protection Act 2018 will be amended to this effect soon. It is something to be considered for the future, hopefully.

Legal professionals working in this field play a crucial role since they are able to discuss these issues with their clients in the process of will drafting and estate planning. Therefore, efforts of professional organisations and associations, such as STEP (an association of professionals working in family planning, including probate and succession laws),³⁰ should be recognised, as they aim to harmonise the practice and raise awareness within the legal profession.³¹

7.3 Concluding Remarks

This book does not aim to encompass all the necessary reforms, changes and challenges in this complex and broad area. Instead, it offers a thorough legal examination of the theoretical and doctrinal issues surrounding online death and digital assets. It discusses and links different areas of the law, humanities and social sciences to provide critical answers and suggestions about what happens to the individual's digital assets and online identity after they die.

²⁹ Recital 27 of Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) OJ EU L 119/1.

³⁰ STEP, 'About us' <<http://www.step.org/about-us>> last accessed 1 August 2021.

³¹ See STEP, 'Digital assets: Practitioner's guide, England and Wales' (2020) <<https://www.step.org/system/files/media/files/2020-03/Digital%20Assets%20Practitioner%20Guide%20-%20England%20&%20Wales.pdf>> last accessed 1 August 2021. For a useful guide by a legal professional, see L Sagar, *The Digital Estate (Wills and Probate)* (Sweet & Maxwell 2018). For very helpful practical guidance with tips and advice, see S Hartung, *Digital Executor(R): Unraveling the New Path for Estate Planning* (FriesenPress 2021).

The book develops the concept of post-mortem (later, postmortal) privacy, that is, privacy of deceased individuals, which I pioneered conceptually in my previous work with Edwards, and later further developed in my research. It offers suggestions for law reform, useful for legislators and policy-makers. It also provides answers of interest for the academic community, the technology industry and the legal profession. Finally, the book provides some practical considerations and advice beneficial for all Internet users in the UK, EU, US and more broadly.